Maui – Tsunami ESF-6 playbook

Data as of 2026-10-28

County: Maui · **Hazard:** Tsunami

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

People at risk **18,239**	Shelter demand (central) **912**	Shelter demand low – high **365 – 2,006**
Limited English (C) **517**	Power-dependent (M) **111**	Disability (I) **1,799**
Age 65+ (I) **3,289**	People without a vehicle (T) **731**	Known 2026 sites usable **5**
At risk, extreme tsunami zone (worst case) **24,391**		

Shelter-seeking rates: low 2%, central 5%, high 11.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

2. Scope: county total vs. one community

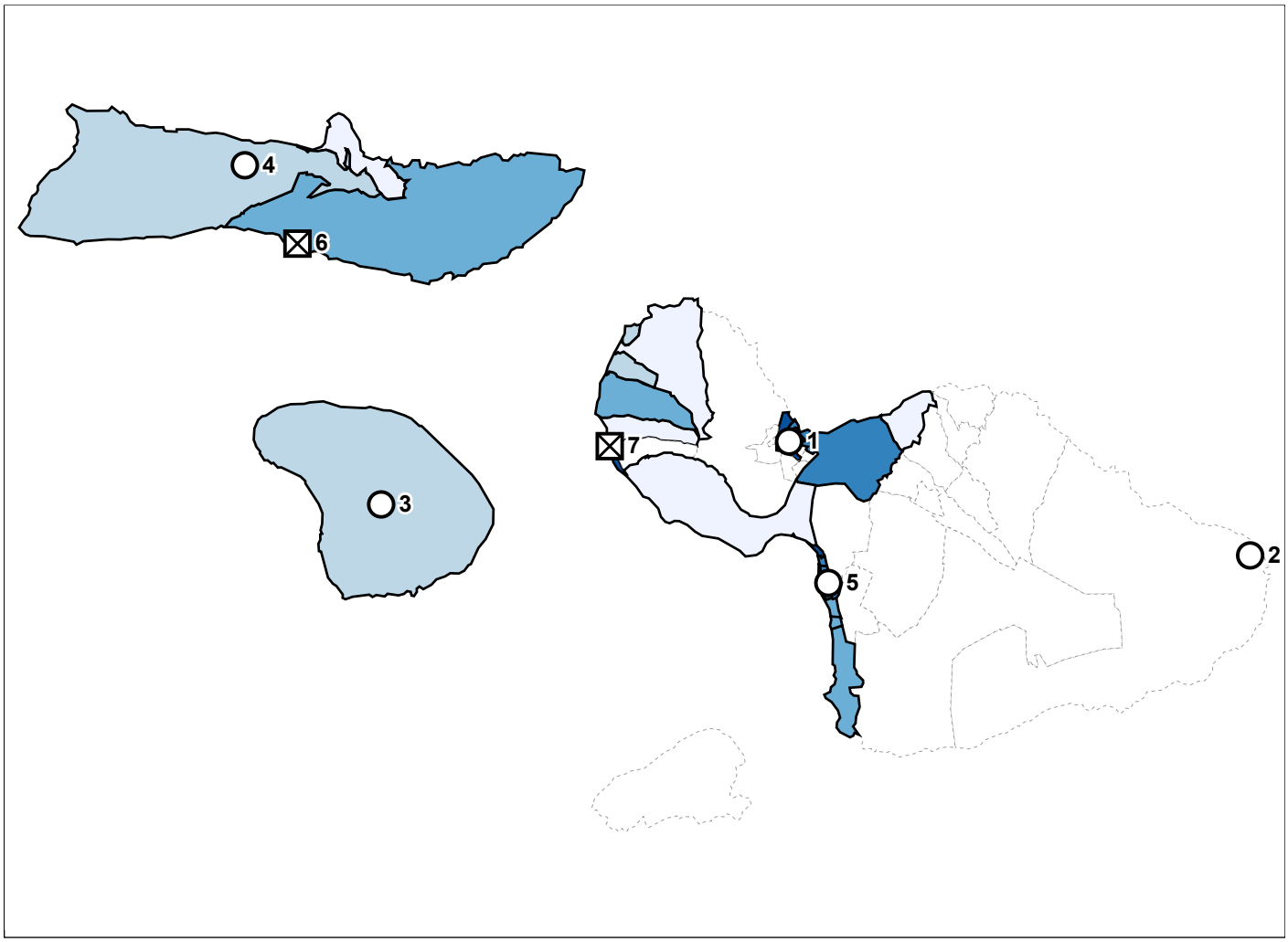
County total = upper bound. It assumes every at-risk tract is hit at once (share inside tsunami zone). Real events usually hit fewer communities.

Largest single community (highest-demand census tracts)

Tract	At risk	Shelter low / central / high	Limited English	Power-dep.	Disability	65+	No vehicle
Tract 307.13 · Kihei	2,557	51 / **128** / 281	22	9	191	259	79
Tract 314.04 · Lahaina	2,213	44 / **111** / 243	69	17	218	495	88
Tract 307.06 · Kihei	1,957	39 / **98** / 215	18	9	245	263	8

Plan a realistic one-community event with the first row.

3. Map: AFN need and known sites



- AFN need 0–14 people (class 1 of 5)
- AFN need 14–38 people (class 2 of 5)
- AFN need 38–138 people (class 3 of 5)
- AFN need 138–253 people (class 4 of 5)
- AFN need 253–392 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

4. Known sites (opened in 2026 events)

#	Status	Site	Address	Reason	Times opened 2026
1	✓ **Usable**	Baldwin High School	1650 Kaahumanu Ave., Wailuku	outside tsunami evacuation zone	1
2	✓ **Usable**	Hāna High & Elementary School Cafeteria	4111 Hāna Highway, Hāna	outside tsunami evacuation zone	1
3	✓ **Usable**	Lānaʻi Youth Center	101-165 Fifth St., Lānaʻi City	outside tsunami evacuation zone	1
4	✓ **Usable**	Molokaʻi Baptist Church	495 Puupeelua Ave., Hoolehua	outside tsunami evacuation zone	1
5	✓ **Usable**	South Maui Community Park Gym	1501 Liloa Drive, Kīhei	outside tsunami evacuation zone	2
6	✕ **Excluded**	Kilohana Recreation Center	344 Kamehameha V Highway, Kaunakakai	inside tsunami evacuation zone	1
7	✕ **Excluded**	Lahaina Civic Center Gym	1840 Honoapiʻilani Highway, Lahaina	inside tsunami evacuation zone	2
8	**Not located**	Mayor Hannibal Tavares Community Center (Social Hall)	91 Pukalani St., Pukalani	not located – not on map	1

Candidate public schools in this county usable for this hazard: **31**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

Need	Quantity	Unit	Proposed lead	Proposed support	Pledged	Gap
Cots / sleeping space	912	cots	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤72 h)	18,239	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Floor space (≤2 weeks)	36,478	sq ft	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Water	912	gal/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Meals	1,824	meals/day	red_cross	dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community	—	confirm with partner
Toilets (≤72 h)	23	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Toilets (≤2 weeks)	46	toilets	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies	—	confirm with partner
Shelter staff	65	staff per shift	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community	—	confirm with partner
Power outlets for power-dependent residents	6	outlets	doh	county_em; health_care; utilities	—	confirm with partner
Accessible transport	731	people in no-vehicle households at risk	county_em	dhs; hiema; dot; business_community; other_agencies	—	confirm with partner
Pet space – dog households	144	households with a dog	county_em	hawaii_voad	—	confirm with partner
Pet space – cat households	84	households with a cat	county_em	hawaii_voad	—	confirm with partner
Children's area	186	children in shelter	county_em	dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy	—	confirm with partner
Interpreters – Ilocano	11	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Tagalog, Filipino	6	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Spanish	2	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Japanese	2	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Marshallese	1	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner
Interpreters – Hawaiian	1	people with limited English in shelter	dhs	county_em; hiema; hspls; doh; oha; faith_based; other_agencies	—	confirm with partner

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Annual pre-positioning review (no official product) County (sets the date)	PRE-POSITION: confirm shelters/refuge outside the tsunami evacuation zone (and extreme zone), transport plan for no-vehicle residents inside the zone, pre-approved multilingual evacuation messages.	County Emergency Management (each county)	hiema, dhs, dot
DURING	Tsunami Warning NOAA Pacific Tsunami Warning Center (PTWC)	Evacuate the zone; open refuge outside the zone; prioritize residents who cannot evacuate on their own.	County Emergency Management (each county)	hiema, dot, red_cross
AFTER	Warning cancelled or all clear County / NWS / PTWC	Shelter displaced households, feeding, power for medical devices, donations and volunteers.	American Red Cross	hawaii_voad, doh, utilities, salvation_army

7. Priority outreach and languages

Top 10 census tracts by AFN need (tract-level counts only)

#	Tract	At risk	Power-dep.	Disability	Limited English	No vehicle	AFN need
1	Tract 314.04 · Lahaina	2,213	17	218	69	88	392
2	Tract 309.03 · Waihee-Waiehu	1,497	10	166	52	75	303
3	Tract 307.13 · Kihei	2,557	9	191	22	79	300
4	Tract 311.06 · Kahului	1,172	7	137	65	89	298
5	Tract 307.06 · Kihei	1,957	9	245	18	8	279
6	Tract 307.12 · Kihei	1,613	12	128	79	35	253
7	Tract 319 · Kahului	815	4	93	75	79	252
8	Tract 307.11 · Kihei	1,434	7	88	36	47	178
9	Tract 309.02 · Wailuku	906	5	67	23	67	162
10	Tract 307.08 · Kihei	959	6	115	27	0	148

Languages: people who speak English less than "very well" (Kauai + Maui)

Kaua'i and Maui share one language list: ACS PUMS reports both counties (and Kalawao) as a single area, PUMA 00100, so they cannot be separated.

Language	People 5+	90% MOE
Ilocano	9,131	±1,439
Tagalog, Filipino	5,074	±926
Spanish	1,711	±554
Japanese	1,339	±417
Marshallese	633	±488
Hawaiian	515	±289

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	0.02 / 0.03 / 0.05	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	0.02 / 0.05 / 0.112	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption.
Shelter-seeking rate – Tsunami	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	0.02 / 0.05 / 0.11	team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	all	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	tsunami_zone_share	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Inland flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
Footprint – Coastal flooding	nri_top_share_by_county (25%)	Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption).
floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
water gal per day	1	Florida DEM (2010)
meals per day	2	Florida DEM (2010): two hot meals a day plus snacks
persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
power outlets per power dependent person	1	team_assumption
persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6

Source	Fetched	URL / file
County shelter activations 2026 (official county releases)	see table	data/sources/county_shelter_activations.csv
Census TIGERweb 2020 Census Designated Places (tract community labels)	2026-10-27	https://tigerweb.geo.census.gov/arcgis/rest/services/TIGERweb/tigerWMS_Census2020/MapServer/28
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement	2026-10-15	HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww
coastal flood warning		https://www.weather.gov/okx/marinedef

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.